

The Question of Origin: The Causal Loop

1 1. The Question of Origin: The Causal Loop

In this premise, the continuity of the project depends on your 60-year-old self returning to convince your 30-year-old self not to give up. However, for the 60-year-old version to be there, the project must have continued, implying that somehow, continuity was guaranteed to complete the causal loop.

If we attempt to answer "what ensures continuity?" in a single timeline, the answer remains suspended: it is *your 60-year-old self*. This cycle creates what is called the *bootstrap paradox* or "originless" phenomenon, since the motivating event for the 30-year-old version depends on the return of the 60-year-old version itself—a cause that sustains itself without a "first motive" outside the cycle.

2 2. Attempt at Retroactive Logic: "It Has Always Been This Way"

One possible interpretation in a single timeline is to assume that this sequence of events "has always been this way" and therefore does not require an initial cause outside the cycle. In this scenario, the event is closed, and there is no "first time" that led your 60-year-old self to return, as the cycle is self-sustaining. This implies that, in a single timeline, you will always find yourself returning at 60 years old to motivate yourself, creating a kind of loop without an external beginning. In other words, your continuity always depends on a "future you," and this return to the past is a fixed and inevitable component in the history of your timeline.

3 3. The Temptation of an Alternative Timeline

Logic, however, often inclines us to think that *there should exist an initial cause outside the cycle* that led your 60-year-old version to complete the project for the first time, allowing it to return and motivate your 30-year-old version. In this case, there seems to be an *implicit suggestion* of an alternative timeline:

- In a possible initial timeline, you decided to continue the project for a different reason (through personal inspiration, external influence, etc.) without interference from your 60-year-old self.

- After being completed, this “primary” timeline would allow the 60-year-old version to return and ensure continuity in the “current” timeline, where you initially gave up.

This hypothesis of distinct timelines would allow for a logical explanation: the *origin* of continuity would come from another cause that ultimately established an alternative causal loop in which returning to the past is necessary to close continuity.

4 4. Rationalization in a Single Timeline: Originless Feedback

Returning to the single timeline model, the initial cause does not exist outside the closed loop. This implies that, *in a single timeline, your 60-year-old self is the very guarantee of the loop*, regardless of an initial motivation. In this context, you are trapped in a cycle where continuity has always been guaranteed by the return to the past of your 60-year-old self. This feedback eliminates the need for a cause outside the cycle.

Therefore, in a single timeline:

- **Your 60-year-old self is, has always been, and will always be the guarantee of the project’s continuity** for the 30-year-old version, and this continuity depends solely on the return to the past.
- There is no “original beginning” where an initial decision led to continuity; the sequence is eternally closed, with each 60-year-old version returning to the past to complete the loop.

This paradox, in a sense, escapes linear causality, and is only possible by assuming that time and causality in this closed loop are cyclically determined and solely dependent on your future intervention, without an initial origin.

Inconsistency and the Reality Paradox

1 1. Inconsistency and the Reality Paradox

In your premise, the 30-year-old version gave up on the project, theoretically meaning it was never materialized. In this scenario, a 60-year-old version of yourself who would return in time to ensure continuity would seem not to exist, as it depends on a future event that was already negated by the act of giving up. In a “causal” and traditional timeline, this would indeed be inconsistent. But if we treat time not as a linear sequence, but as a totality where the future, present, and past have a structural equivalence, we can approach this issue from a different perspective.

2 2. Invariant Structure and the Totality of Time

You suggest that what maintains continuity is something beyond the individual experience and specific decisions of each version (30, 40, or 60 years old), something that resembles a “totalizing information” or an invariant structure. This structure would be timeless and self-consistent, functioning as a *topological morphism* between the past, present, and future. In this context:

- **Equivalence between Temporal Surfaces:** Imagine that the past, present, and future are “topological surfaces” equivalent to each other, each expressing the totality of the process in itself. This view implies that, for any point in time, the other surfaces (past and future) are equally real and interconnected. Each moment carries within itself an equivalence with the “whole,” and this equivalence keeps the cycle complete and cohesive.
- **Invariance of Information:** The continuity of the project, then, would not be based solely on the decisions of the 30- or 60-year-old versions but on a *fundamental information* that permeates and links these surfaces. This non-local information field (not restricted to a moment in time) exists as a self-continuity structure that guarantees the realization of the project in any “version” of time.

3 3. Self-Continuity as Inherent Will

This idea of self-continuity can be seen as a sort of “will” the project has to self-realize. It is not that the 30-year-old version needs to make a decision or that the 60-year-old needs to ensure continuity; it is the very *structure of the cycle* that manifests through these ages, like an instinct to self-complete.

This self-continuity or “will” to materialize approaches an almost essentialist concept, where the project exists as a “Platonic” idea that transcends time, being both the cause and effect of its own existence. In the end, the materialization of the project at 40 occurs because it has “always been there” in an invariant and intangible way, independent of momentary decisions.

4 4. Temporal Equivalence and Totalizing Structure

With the topological view of equivalence you suggested, the past, present, and future become morphisms of a single structure. In other words:

- **The Totality Contained within the Parts:** Each moment (past, present, future) contains within it the totality of time, implying that the “complete history” of the project is integrated within each instant. Thus, the 30-year-old version’s giving up is, paradoxically, part of the project’s continuity, as it exists only as a surface of the total process.
- **The Structure as Temporal Unity:** This unified structure implies that time travel is not so much a “modification” of a state of giving up, but rather the natural expression of a temporal cycle that is not bound to a linear sequence. Thus, the return of the 60-year-old version is not an alteration of a past event, but the manifestation of a state of self-continuity contained in all ages and moments.

5 Conclusion

From this perspective, what guarantees the continuity of the project is the very invariance of the temporal structure, where each “version” of you (30, 40, 60 years old) is part of a whole that already contains the totality of the process. Instead of there being a “first impulse” for continuity, the project is self-sustained by an equivalence among all parts of time.

Thus, time travel is not a correction of a broken line, but a natural expression of the topological structure of time, where past, present, and future are different surfaces of the same “totality.”

General Relativity, Topology, and Temporal Philosophy

1 1. General Relativity and the Topology of Spacetime

In Einstein's general relativity, time and space are interdependent and form a "fabric" of spacetime, which can be distorted by mass and energy. Some solutions to relativity, such as Kurt Gödel's, allow for closed timelike curves (CTCs), where a particle could theoretically return to its own past. This provides a theoretical foundation for closed causal cycles, where the conditions of the "past" and "future" can mutually sustain each other without a clear initial cause. Additionally, in relativity, equivalent surfaces (as you described) could be viewed as different "slices" of spacetime that share a relationship, maintaining a structural invariance. This connects well with your vision of equivalence between past, present, and future.

2 2. Information Theory and the Hypothesis of Non-Local Information

The hypothesis that information may exist outside a specific point in spacetime is explored, for example, in quantum systems theory and the holographic principle. This principle suggests that all information about a volume of space can be described on its surface, implying that time, causality, and even the contents of a region can be "stored" or equivalent to a broader structure. The idea that the "instinct of self-continuity" of the project is not tied to an initial cause but to the atemporal structure of information aligns with the view that time and space may essentially be manifestations of an underlying non-local information.

3 3. The Many-Worlds Interpretation and Alternative Timelines

While you are focusing on a single timeline, the many-worlds interpretation (or multiverse) of quantum mechanics provides an interesting alternative to think about self-consistency. If each potential decision generates an alternative

timeline, then theoretically there could exist an “original” timeline where the project was independently continued and that influences the timeline you are in now. In your view, the “instinct of self-continuity” could be considered as a kind of *morphism* between these timelines, a structural connection that transcends individual choices and maintains the unity of the project in totality.

4 4. Topology Theory and Homotopy Classes

In topology, homotopy classes represent paths or equivalent surfaces that can be continuously deformed into one another. Similarly, we could consider each point (ages 30, 40, 60) as a “surface” that, in topological terms, is equivalent to the others, forming a single structure. This supports your idea of equivalence between past, present, and future, suggesting that time is a “topological totality” where each age represents a surface containing the whole. Thus, your “self” at each age is a manifestation of this totality, and the project can be realized as a structural property of this topology, rather than a simple sequence of causes and effects.

5 5. Temporality and the Philosophy of Time

Philosophers such as Henri Bergson and J.M.E. McTaggart explored the idea that time could be a totalizing structure rather than a linear one. Bergson, for example, describes time as a continuous and indivisible duration, while McTaggart suggests that time might be an illusion in our perception, a phenomenon that is, in essence, merely an “appearance” of reality.

Implicate Order, Holomovement, and Non-Local Information

1 1. Implicate and Explicate Order

Bohm proposed that the universe possesses two orders: an *implicate order*, which is fundamental and contains all information in a “folded” and unmanifest state, and an *explicate order*, which is the visible or “unfolded” manifestation of the reality we experience. In your analogy, where past, present, and future are “topological surfaces” that are equivalent and interconnected, we find a similarity to Bohm’s implicate order. The continuity of the project would be a manifestation of the implicate order — a totality where all possibilities already coexist in a unified form, and the interaction with the versions at 30, 40, and 60 years would be an unfolded expression of this totality, the explicate order.

2 2. Holomovement and the Totalizing Nature of Reality

Bohm uses the term *holomovement* to describe the dynamic and continuous movement of the implicate order, which is in constant flux and transformation. This concept suggests that everything in the universe is interconnected in a holistic movement. Your vision of a self-continuity or an “inherent will” of the project to self-realize can be seen as a manifestation of this holomovement, where the project is not just a series of isolated events, but an expression of a constantly changing totality. What you refer to as the “instinct of self-continuity” aligns with the idea that the fundamental information of the project is flowing continuously in the holomovement, connecting and synchronizing all ages and versions of yourself.

3 3. Information as Non-Local Totality

Bohm also argued that reality is non-local and that the separation between parts of the universe is an illusion. All information is present in every point of the universe, as in a sort of “holography” of the totality. By considering past, present, and future as topologically equivalent, and that the project’s continuity is guaranteed not by linear events but by a non-local informational structure,

you closely approach Bohm's idea of a non-local order. For him, this non-local order implies that the whole is present in each part, just as, in your model, the totality of the project would be "contained" and accessible in each of the temporal surfaces (30, 40, 60 years).

4 4. The Holographic Metaphor and the Project as Totality

Bohm used the metaphor of the hologram, where each part of a hologram contains the information of the whole, to explain how reality might be a holistic expression, where each fragment of spacetime reflects the totality. This holographic view perfectly fits with your idea that time as a whole is contained within each of the ages, and the project itself is a manifestation of this totalizing "will." The project is not realized merely because you, at ages 30 or 60, make a decision, but because it is a "pattern" contained within the total structure of spacetime — or, in Bohm's words, within the implicate order.

5 5. Non-Linear Causality and Totality Feedback

Bohm was critical of strict linear causality and suggested that reality operates through non-linear interactions and feedback from a broader totality. Your suggestion that the "instinct" for the project's continuity lies not in a specific initial cause but in the atemporal structure of the project itself also aligns with Bohm's critique. This non-linear causality implies that the feedback from your 60-year-old self to your 30-year-old self is a form of "holistic causality," where each version of yourself acts as part of an interdependent totality. This non-linear causality allows the return to the past to be seen not as a paradox, but as an expression of the connected totality, where cause and effect are aspects of a single, indivisible structure.

Temporal Totality Structure and Project Continuity

1 1. Temporal Totality Structure

We define time T as an equivalence structure between specific moments $t \in T$, where past, present, and future are interconnected and equivalent. We denote the structural totality as a continuous function $F : T \rightarrow \mathbb{R}^n$ that maps each point t to an information surface associated with the project.

$$F(t) \approx F(t') \quad \forall t, t' \in T$$

2 2. Self-Continuity and Topological Equivalence

We define the topological equivalence between the moments t_{30} , t_{40} , and t_{60} (representing ages 30, 40, and 60, respectively) using homotopy classes. Let H be the homotopy class defining the structural equivalence among these moments:

$$[t_{30}] = [t_{40}] = [t_{60}] = H$$

3 3. Non-Local Connection and Informational State

We define a global informational state Ψ as a state dependent on H , representing the project's totality. We use a projection operator P_t to extract the manifestation of Ψ at each moment t :

$$P_t(\Psi) = \psi_t$$

where ψ_t represents the local information at time t , which is a partial expression of Ψ .

4 4. Holomovement and Temporal Continuity Flow

We introduce a flow operator D , representing Bohm's holomovement, which ensures temporal continuity and the "will" for the project's self-continuity:

$$D\psi_t = \frac{\partial\psi_t}{\partial t} = 0$$

The equation above implies that the variation of the local information ψ_t over time is zero, reflecting the invariance of the structure.

5 5. Holographic Structure and Non-Local Information

We define the totality structure Ω as an integral of the global information Ψ over the equivalence surface H :

$$\Omega = \int_H \Psi dH$$

Since Ω is constant and includes all temporal manifestations ψ_t , each equivalent surface in T retains the totality of the project's information.

6 6. Conditionality of Temporal Continuity

Finally, to ensure that self-continuity is a property of the global structure Ω and not of individual events, we use a self-consistency condition relating each moment t to the totality Ω :

$$F(t) = \Omega \quad \forall t \in T$$